Field Training 1

SAS Visual Analytics

Report

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ID:2022513643 / Intelligent Systems

Instructors:

Dr. Mohamed Mostafa Abbassy

Contents

Introduction

This report presents an overview of our journey through two comprehensive courses on SAS Visual Analytics for SAS Viya, culminating in a hands-on project that allowed us to apply the knowledge we gained.
The first course served as our steppingstone, introducing us to the fundamentals of data preparation, data discovery, and report creation. We learned to organize content in SAS Drive, interact with reports using the SAS Report Viewer, access and investigate data, prepare data using SAS Data Studio, perform data discovery and analysis, and create interactive reports.
Building on this foundation, the second course propelled us into the advanced realm of data preparation, analytics, and report creation. We explored automated explanation, restructured data for several types of analytics (including geographic analysis, forecasting, network analysis, path analysis, and text analytics) using SAS Data Studio, performed advanced analyses, and created advanced data items, filters, and interactive reports with parameters.
The culmination of our learning journey was a project where we applied these skills in a real-world context, truly embodying the role of business analysts using the advanced functionality of SAS Visual Analytics. This hands-on experience not only reinforced our learning but also provided us with practical insights into the power and potential of SAS Visual Analytics in driving effective business analysis.

Training Details

Training Name:
SAS Visual Analytics

Training Place:
Held online at the Faculty of Computers and Data Science, Alexandria University.

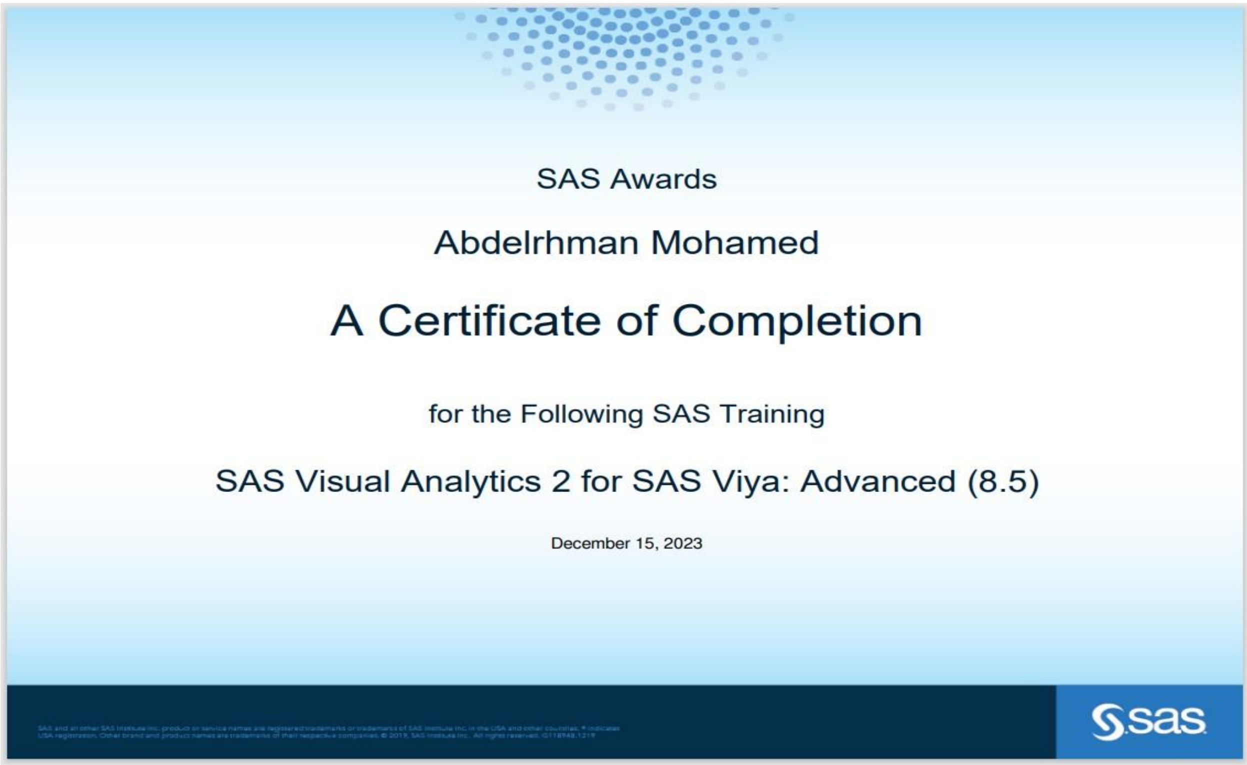
Training Duration:
From October 30th, 2023, to December 5th, 2023 - 36 days

Frequency:
SAS Visual Analytics 1 for SAS Viya: Basics (8.5) – 2 weeks
SAS Visual Analytics 2 for SAS Viya: Advanced (8.5) – 3 weeks

Team members

Name	Department	ID	Tasks
Ahmed Mostafa AbdelRahman	Intelligent System	20221372883	Access Investigate
Abdelrhman Mohamed Abdelhady Hodib	Intelligent System	2022513643	Prepare GARAGE AREA AND LOT AREA ON HOUSE PRICE Report
Ahmed Ehab El-Sayed	Intelligent System	20221445838	HOME AGE AND SALEPRICE HEATING QC AND CENTRAL AIR ON SALE PRICE
Mazen Gaber Mahmoud		20221372110	GROUND LIVING AREA AND SALEPRICE SEASONS AND SALEPRICE

Certificates



SAS Awards

Abdelrhman Mohamed

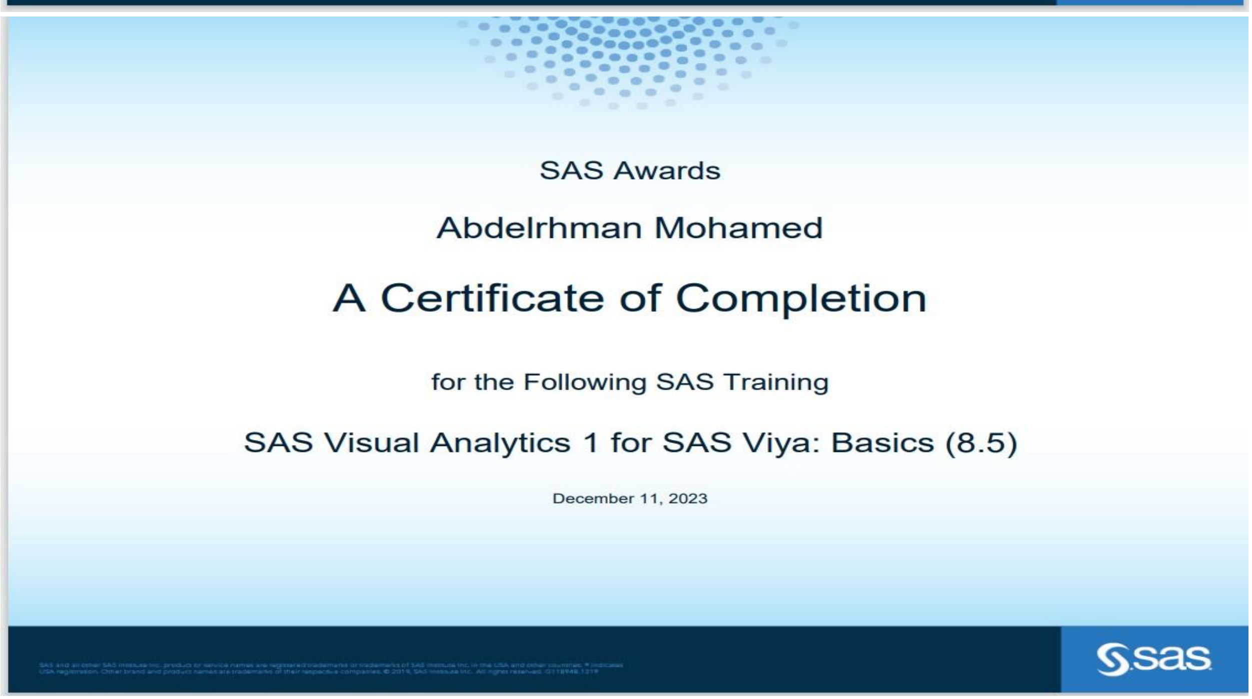
A Certificate of Completion

for the Following SAS Training

SAS Visual Analytics 2 for SAS Viya: Advanced (8.5)

December 15, 2023

sas



SAS Awards

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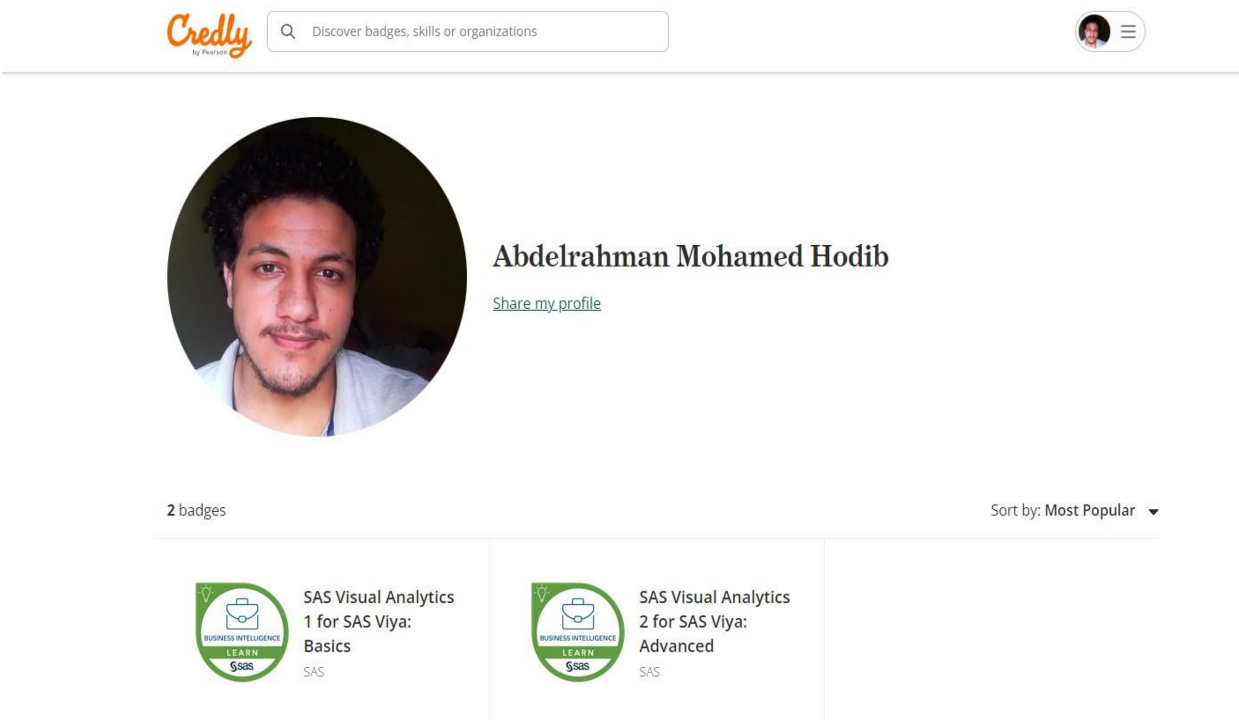
SAS Visual Analytics 1 for SAS Viya: Basics (8.5)

December 11, 2023

sas



Training subjects

SAS Visual Analytics 1 for SAS Viya: Basics (8.5)

1. **Introduction to SAS Visual Analytics**
   This section introduces SAS Visual Analytics and explores the course environment. It includes demonstrations on exploring SAS Drive and viewing reports. The section concludes with a practice session and a review of the solutions.
2. **Investigating Data in SAS Visual Analytics**
   This section focuses on accessing and investigating data. It also covers transforming data using SAS Data Studio, with a demonstration on preparing data. The section ends with a practice session and a review of the solutions.

3. **Working with Data Items**
   This section delves into creating and working with data items. It includes demonstrations on exploring data with charts and graphs and creating data items. The section also covers applying filters to the data.
4. **Designing Reports with SAS Visual Analytics**
   This lesson focuses on creating reports. It starts with creating a simple report, followed by creating interactive reports. The interactive reports section covers working with pages and ranks, prompts and actions, and hidden pages and page links. The lesson also includes a section on working with display rules, specifically graph-level display rules. Each topic is accompanied by a demonstration and a practice session, and the lesson concludes with a review of the solutions.
5. **Analyzing Data in SAS Visual Analytics**
   - In this section, I learnt how to perform data discovery and analysis using SAS Visual Analytics.
   - This involves working with data items, exploring data with charts and graphs, creating data items, applying filters, and performing data analysis.
   - I learnt how to use different types of charts and graphs, understand the data they represent, and interpret the results.

6. **Creating Interactive Reports**
   - The course concludes with a section on creating interactive reports using SAS Visual Analytics.
   - I learnt how to create simple reports, create interactive reports, and work with display rules.
   - This involves understanding how to select the appropriate visualization for my data, customize the appearance of my reports, and add interactivity to allow users to explore the dat

SAS Visual Analytics 2 for SAS Viya: Advanced (8.5)

Overview of SAS Visual Analytics The advanced SAS Visual Analytics 2 for SAS Viya (8.5) course encompasses the following key areas:

1. **Data Restructuring for Geographic Mapping:**

   This module introduces SAS Data Studio and imparts knowledge on how to restructure data for geographic mapping and analysis.

2. **Data Restructuring for Forecasting:**

   This segment of the course educates on how to restructure data for forecasting and the process of performing forecasting.

3. **Network Analysis:**

   This section elucidates the process of restructuring data for network analysis and the creation of a network analysis object.

4. **Path Analysis:**

   This part of the course focuses on the analysis of a path analysis data source.

5. **Text Analytics:**

   This module delves into the analysis of a text analysis data source.

6. **Creation of Advanced Data Items:**

   This segment of the course imparts knowledge on the creation of calculated items and aggregated measures.

7. **Creation of Advanced Filters:**

    This section provides instruction on the creation of advanced filters and advanced interactive filters.

8. **Usage of Parameters for Advanced Reports:**

    This part of the course educates on the usage of numeric parameters, character parameters, and date parameters for the creation of advanced reports.

The various sections of the course collectively provide a comprehensive understanding of SAS Visual Analytics, arming learners with the necessary skills to implement these concepts in real-world scenarios.

Throughout the course, I had the opportunity to engage in practical exercises that allowed me to apply my learnings and solidify my grasp of the concepts. This hands-on experience was instrumental in enhancing my understanding of SAS Visual Analytics.

Upon completion of the course, I had developed a robust understanding of how to utilize SAS Visual Analytics for advanced data preparation, analytics, and report creation. I am now proficient in using automated explanation in SAS Visual Analytics, restructuring data for various types of analytics (such as geographic analysis, forecasting, network analysis, path analysis, text analytics) using SAS Data Studio, performing advanced analyses using SAS Visual Analytics, creating advanced data items and filters, and creating advanced interactive reports with parameters. This knowledge equips me to effectively leverage the capabilities of SAS Visual Analytics in a business context.

Throughout the SAS Visual Analytics 1 for SAS Viya: Basics (8.5) and SAS Visual Analytics 2 for SAS Viya: Advanced (8.5) courses, I acquired a broad spectrum of skills and techniques vital for data analysis and visualization.

1. I became proficient in navigating the SAS Visual Analytics interface and organizing content in SAS Drive.
2. I learned to view and interact with reports, which expanded my understanding of SAS Visual Analytics' capabilities and inspired my own analytical endeavors.
3. I gained experience in accessing and investigating data in SAS Visual Analytics, which involved examining variable distributions, identifying missing data, and spotting interesting patterns or outliers.
4. I learned to prepare data using SAS Data Studio, which encompassed data cleaning, variable creation, and categorical variable encoding. This process taught me how to transform data to make it more amenable to analysis and visualization.
5. I learned to perform data discovery and analysis using SAS Visual Analytics. This involved creating visual representations of data through charts and graphs, applying filters to concentrate on specific data segments, and using automated explanation to highlight key relationships and outliers.
6. I learned to create interactive reports using SAS Visual Analytics, adding interactivity to the reports such as data filtering, detail drilling, and data exploration.
7. Lastly, I learned to perform advanced analyses using SAS Visual Analytics. This included geographic analysis, forecasting, network analysis, path analysis, and text analytics. I also learned to create advanced data items and filters, and to use parameters to create advanced interactive reports.

These learnings have equipped me with a comprehensive understanding of SAS Visual Analytics, preparing me to effectively apply these skills in real-world scenarios.

Final project

Introduction

In this Project we will be applying the SAS Visual Analytics Methodology on the AMES_HOUSING dataset on SAS Drive

SAS Visual Analytics Methodology

The Visual Analytics methodology is a step-by-step process that we can follow when using Visual Analytics. The methodology is divided into five phases: Access, Investigate, Prepare, Analyze, and Report.

1- Access phase

In the Access phase, we identify the analysis tables that will be used in Visual Analytics and load those tables into CAS.

Accessing the AMES_HOUSING dataset, it consists of 10 Columns and 300 Rows
Here is the breakdown of the variables:

- **Lot_Area**: The area of the lot that the house is on.
- **House_Style**: The style of the house (e.g., 1Story, 2Story, 1.5Fin, SLvl, SFoyer, 1.5Unf, 2.5Unf).
- **Heating_QC**: The heating quality and condition (e.g., Ex for Excellent, Gd for Good, TA for Average/Typical, Fa for Fair).
- **Central_Air**: Whether the house has central air conditioning (Y for Yes, N for No).
- **Gr_Liv_Area**: The above-ground living area in square feet.
- **Fireplaces**: The number of fireplaces in the house.
- **Garage_Area**: The size of the garage in square feet.
- **SalePrice**: The sale price of the house.
- **Age_Sold**: The age of the house at the time it was sold.
- **Season_Sold**: The season when the house was sold (e.g., 1 for Winter, 2 for Spring, 3 for Summer, 4 for Fall).

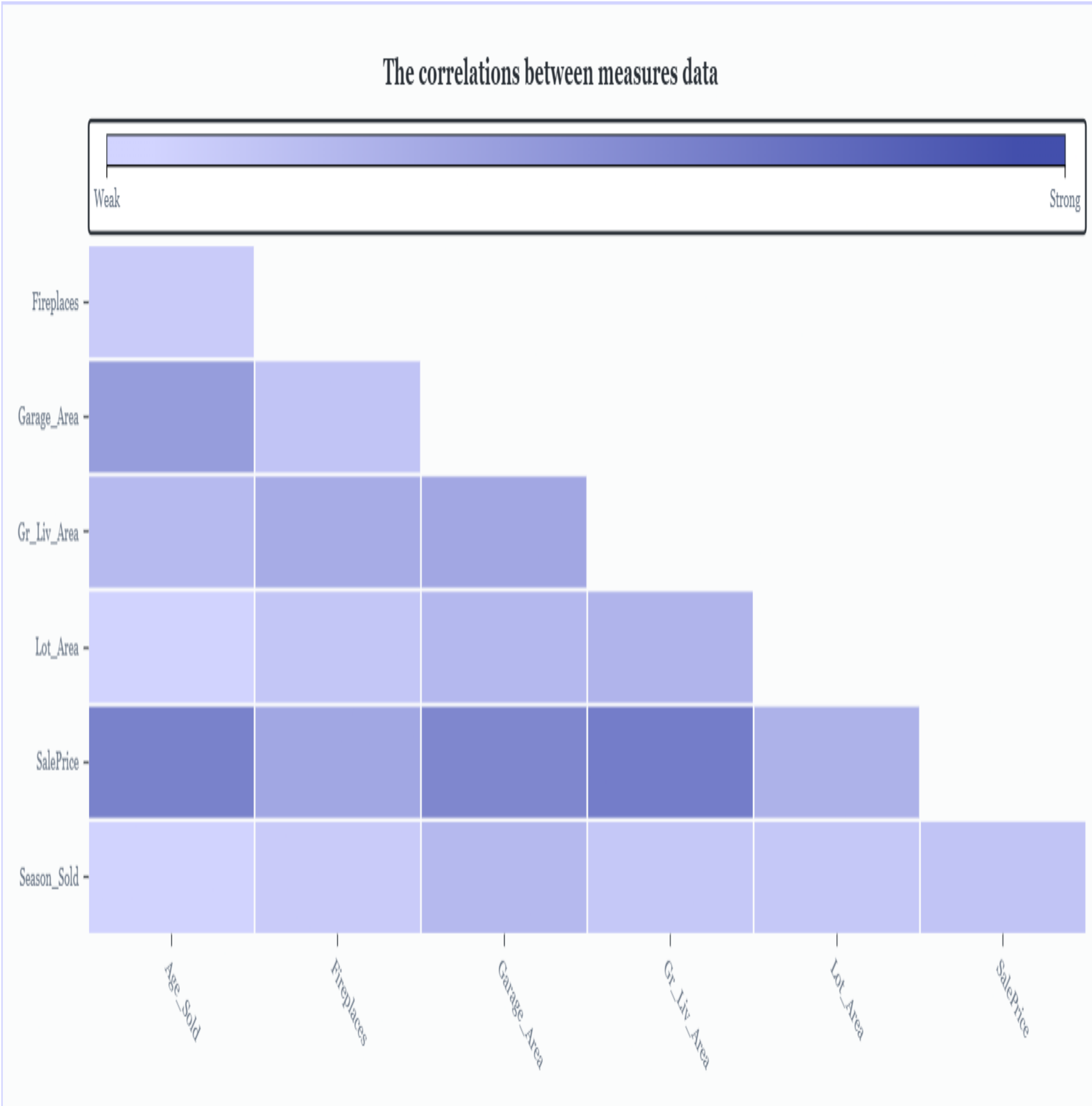
2- Investigate phase

In the Investigate phase, we inspect the tables to determine whether any changes are needed for data items due to data inconsistencies or data quality issues, and we also identify any new data items that need to be created.

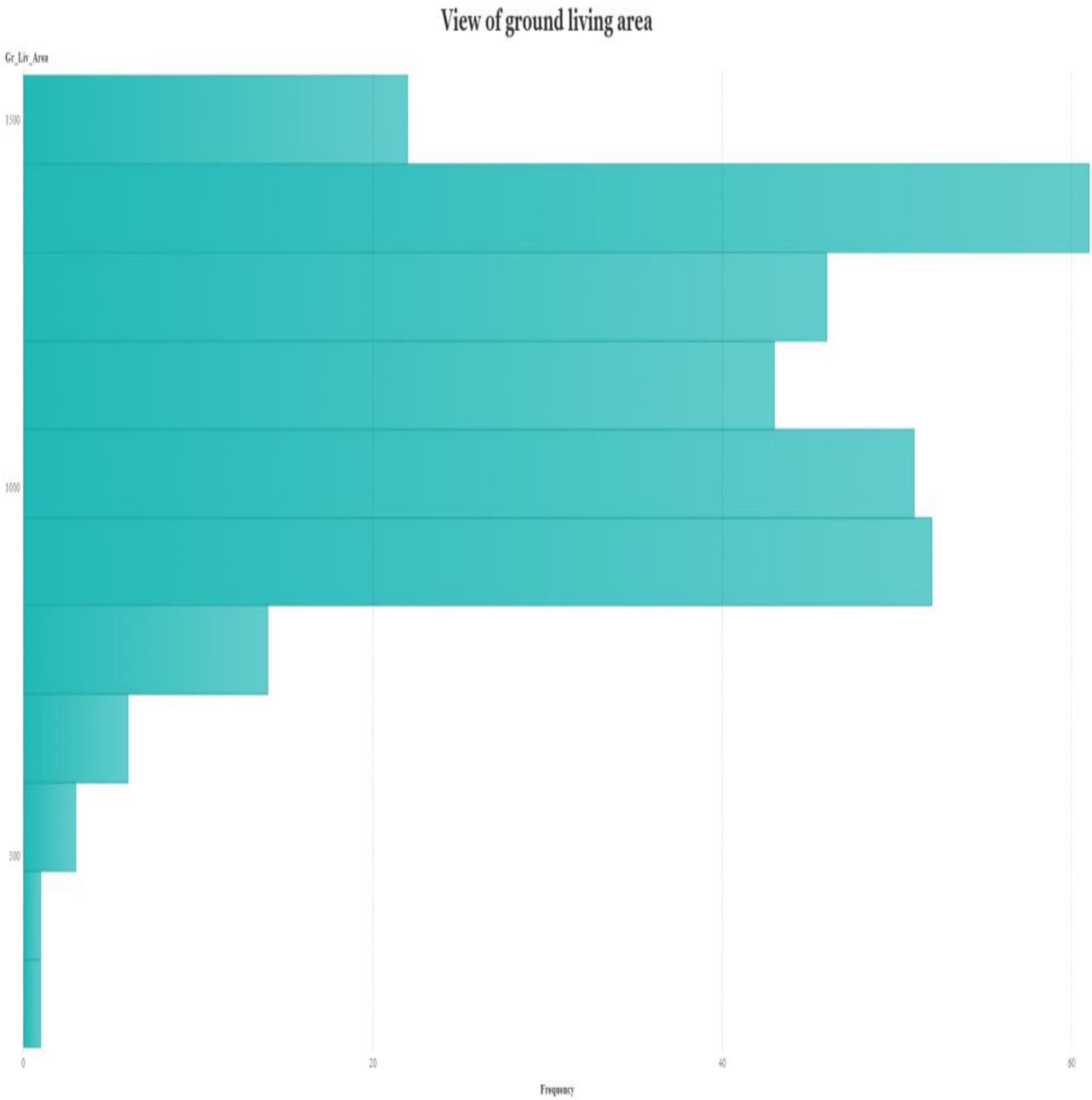
1. We deduced that our data is clean and there are no null values in it.

Column	Unique	Null	Blank	Pattern Count	Mean	Medi
Age_Sold	31.33% (...				45.89	45.
Central_Air	0.67% (2)			1		
Fireplaces	1.00% (3)				0.39	0.
Garage_Area	43.33...				369.45	390.
Gr_Liv_Area	75.67% (2...				1,130.74	1,135.
Heating_QC	1.33% (4)			2		
House_Style	2.33% (7)			4		
Lot_Area	83.33% (250)				8,294.14	8,265.
SalePrice	65.33% ...				137,5...	135,0.
Season_Sold	1.33% (4)				2.55	3.

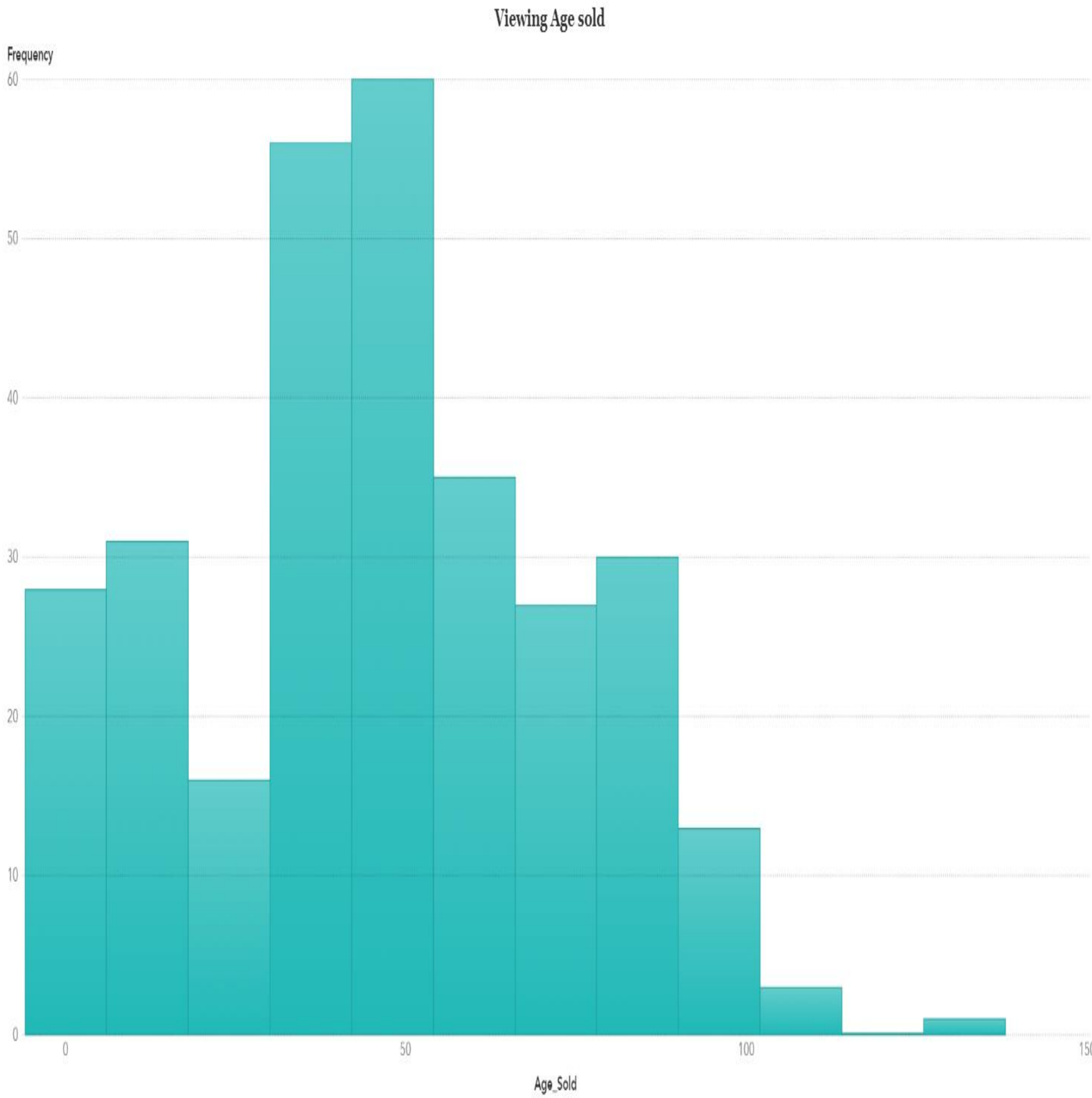
2. We deduced from the correlation matrix between all measures that there is a strong relationship between saleprices and (ground living a rea, Age sold).



3. We can add new category of ground living area of three classes (less than 500,from 500 to 1000, from 1000 to 1500).



4. We deduced that we can add new data category from the age sold measure that ranges from 0 to 135 into 4 classes (less than 10, from 10 to 30, from 30 to 50 , More than 50).



3- Prepare phase

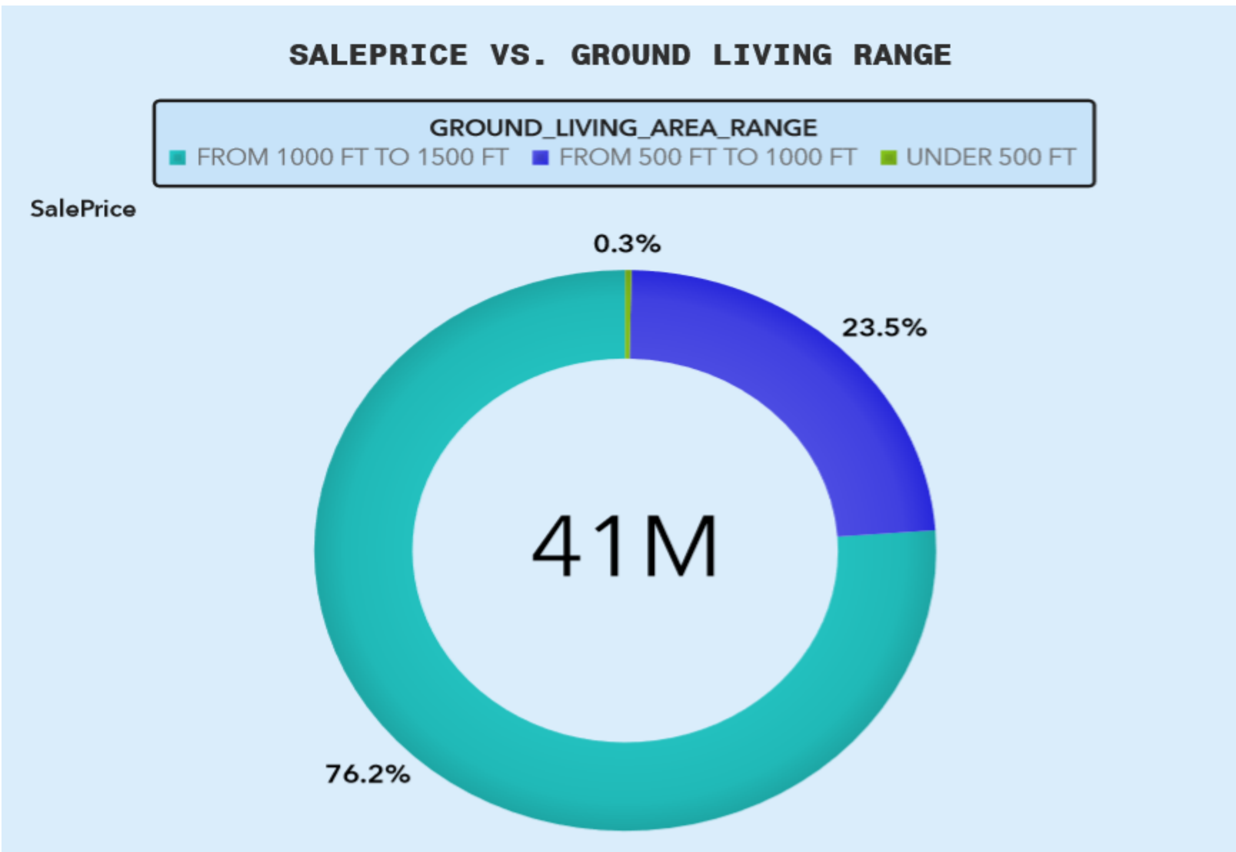
In the Prepare phase, we correct any data quality issues and create any new calculated items needed for analysis.

- We deduced that we can add new data category from the saleprice that has 3 classes (under 100,000, between 100,000 and 200,000, over 200,000).
- We deduced that we can change the measure (season_sold) from (1,2,3,4) to (winter,spring,summer,fall).
- We can convert heating quality condition from (ex,fa,…) to Numbers.

4- Analyze phase

In the Analyze phase, we explore the data to identify any patterns, relationships, and trends.
Now, we'll explore our data with charts and graphs to help us uncover hidden relationships.

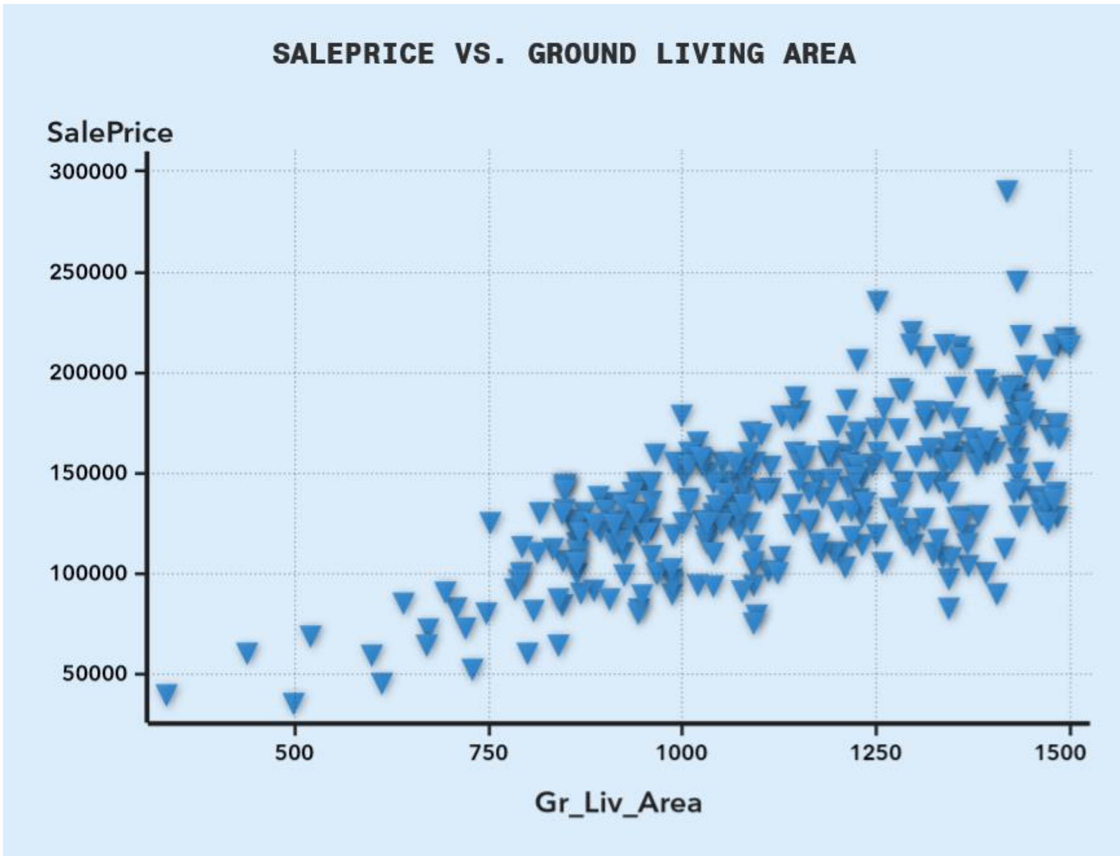
GROUND LIVING AREA AND SALEPRICE



The pie chart shows the distribution of sale prices for the ranges of the ground living area.

- Houses with a ground living area range of 1000 sq ft to 1500 sq ft make up the largest portion , at 76.2%.
- The next largest group is houses with a ground living area range of 500 sq ft to 1000 sq ft, at 23.5%.

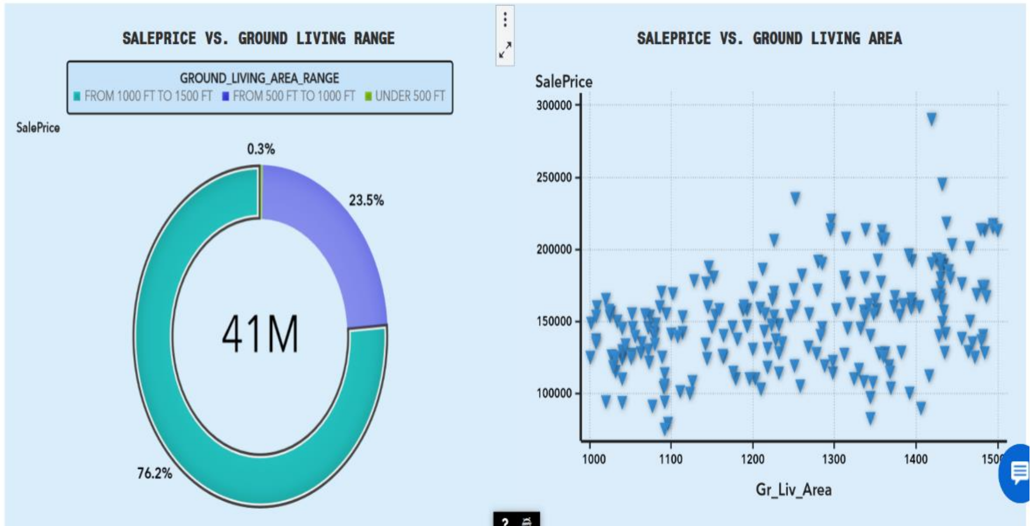
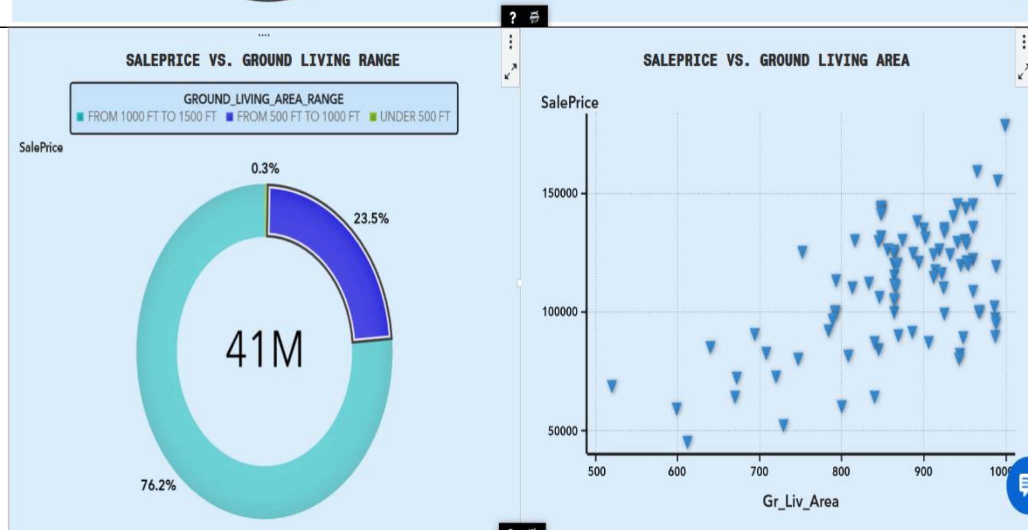
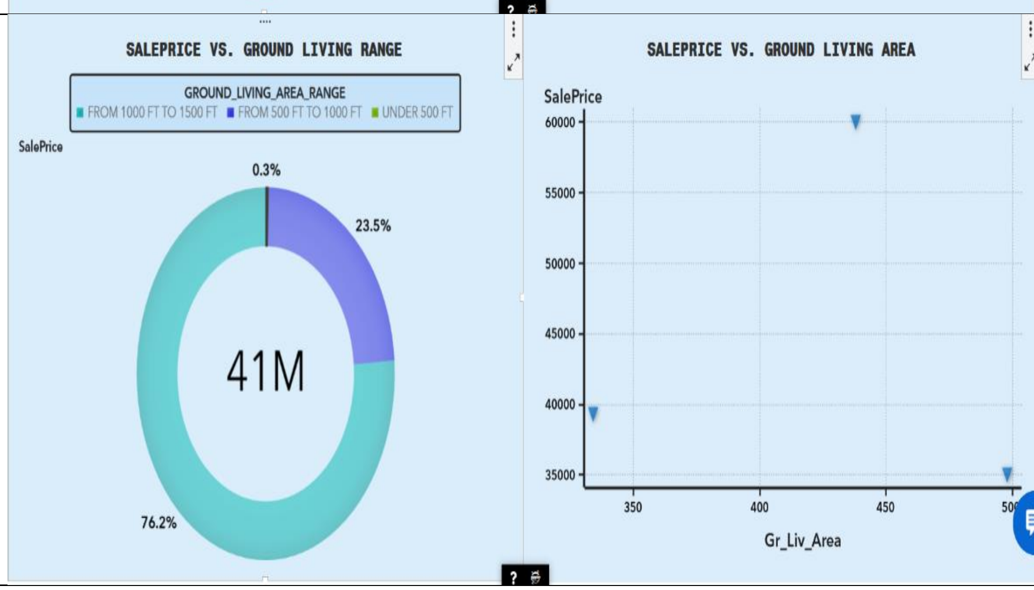
➢ The smallest group is houses with a ground living area under 500 sq ft, at 0.3%.



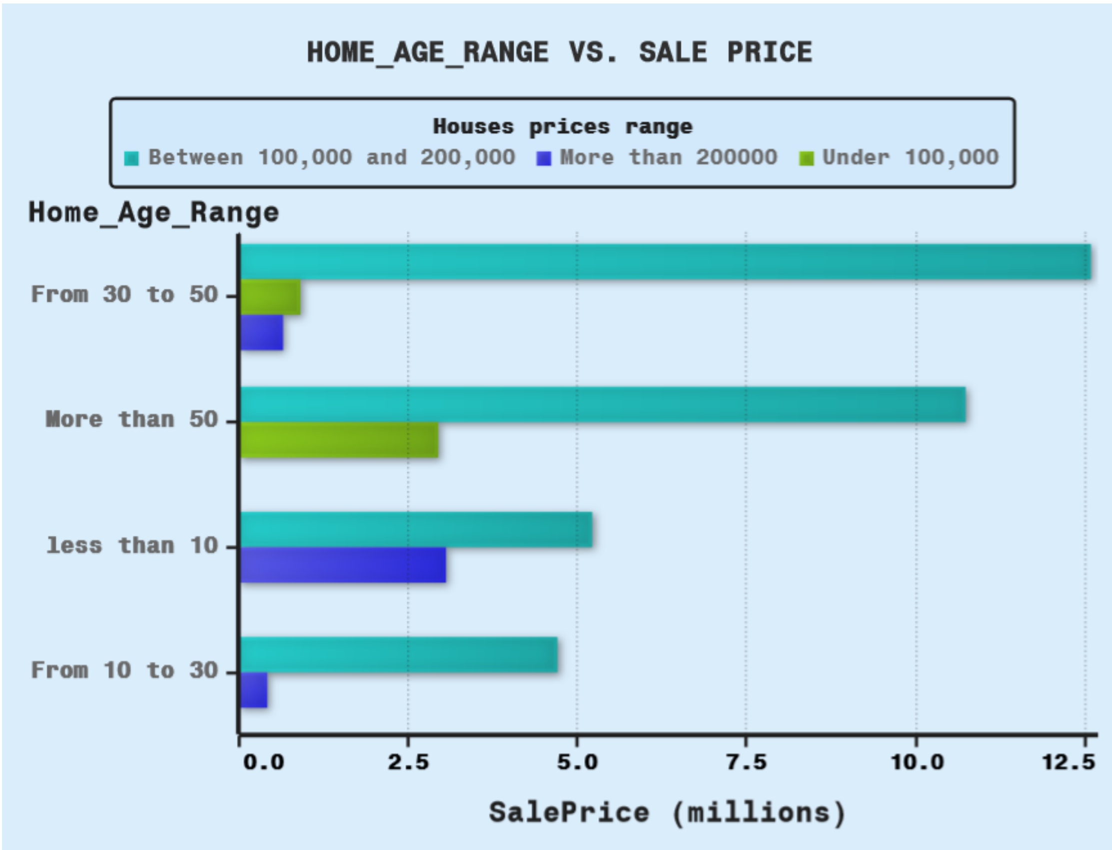
The scatter plot shows the difference in sale price for ground living area of houses.

- The general trend of the graph is that houses with larger ground living areas tend to have higher sale prices. This is because more space is generally seen as a desirable feature in a house, and buyers are willing to pay more for it. However, there is also a lot of variation in the data, and there are many houses that fall below or above the general trend.

Here is the view of sale prices regarding each range of the ground living area

FROM 1000 FT TO 1500 FT	
FROM 500 FT TO 1000 FT	
UNDER 500 FT	

HOME AGE AND SALEPRICE

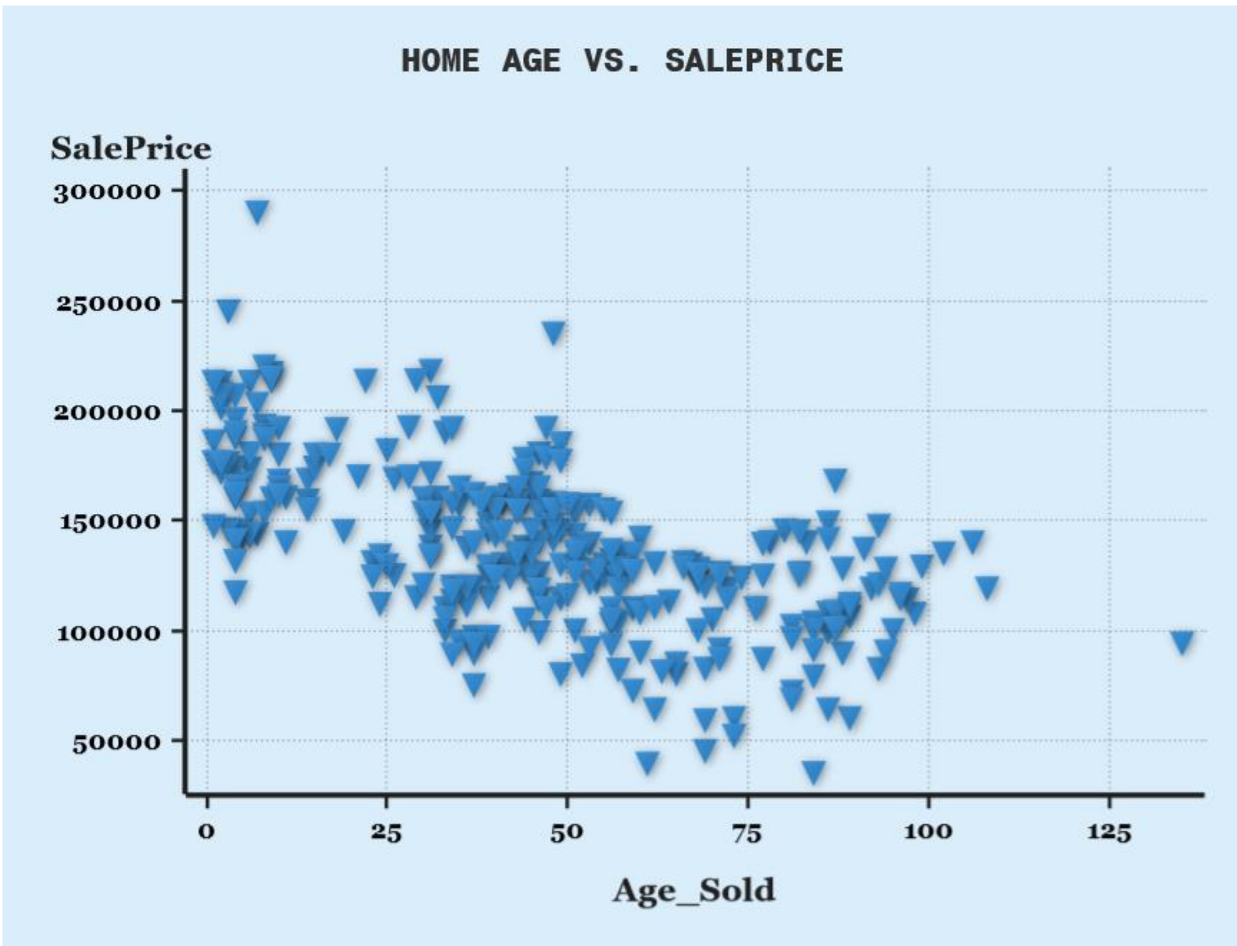


The bar plot shows the sale price of houses by age range.

The graph shows that houses between 30 and 50 years old have the highest median of all saleprice, at around $7.5 million. Houses from 10 to 30 years old have the lowest median of all sale price, at around $2.5 million. Houses more than 50 years old have a median of all sale price of around $5 million.

There are a few possible explanations for this pattern.

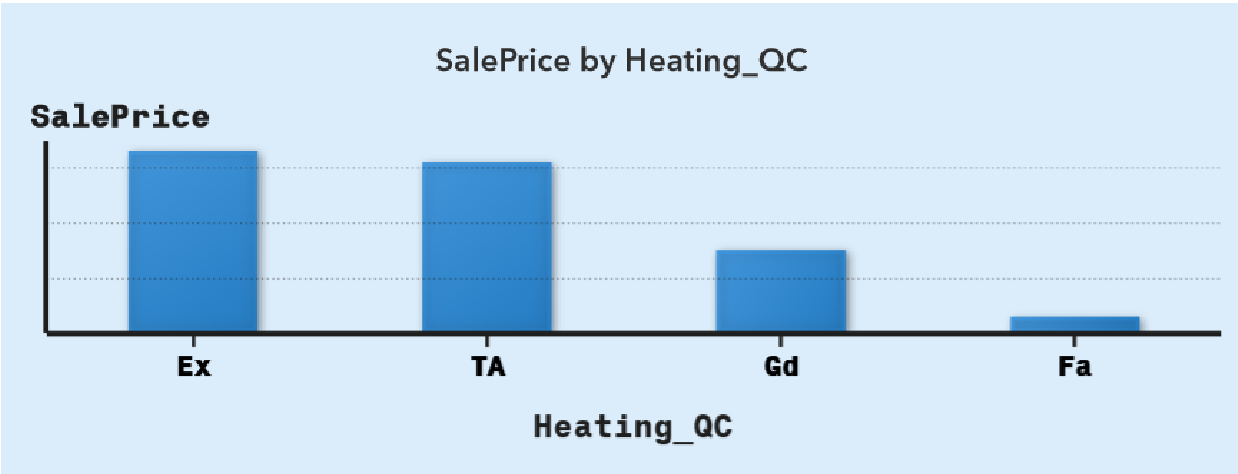
- One possibility is that newer houses are more expensive because they are built with newer materials and technologies.
- Another possibility is that older houses are more sold because they are located in desirable neighborhoods. It is also possible that there is a combination of these factors at play.



The scatter plot shows the distribution of sale price of homes by the age of the homes were sold.

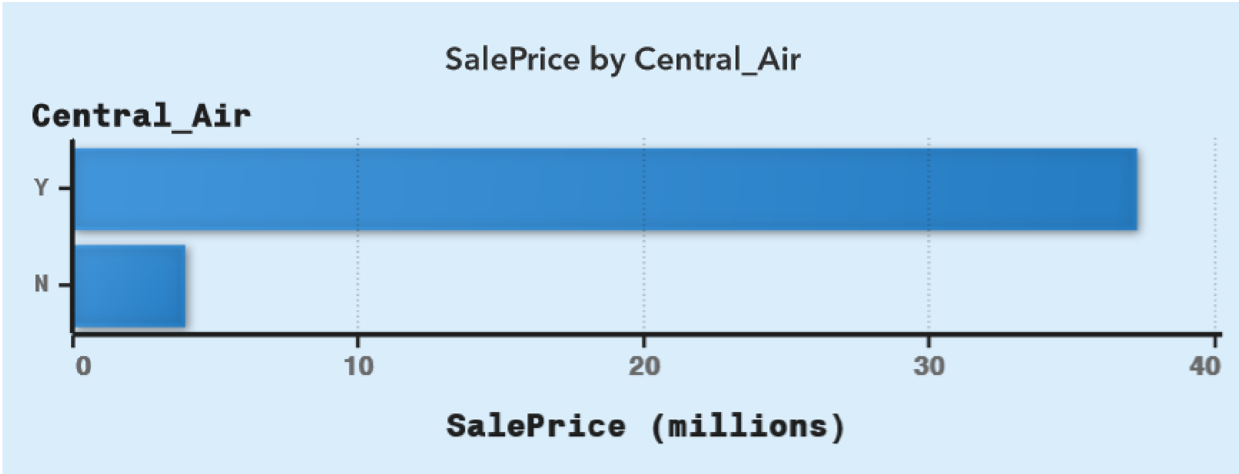
- The graph shows that, in general, newer homes tend to have higher salseprice for more than older homes.

HEATING QC AND CENTRAL AIR ON SALE PRICE



The bar plot shows sales price of homes, by heating quality condition (heating_qc)

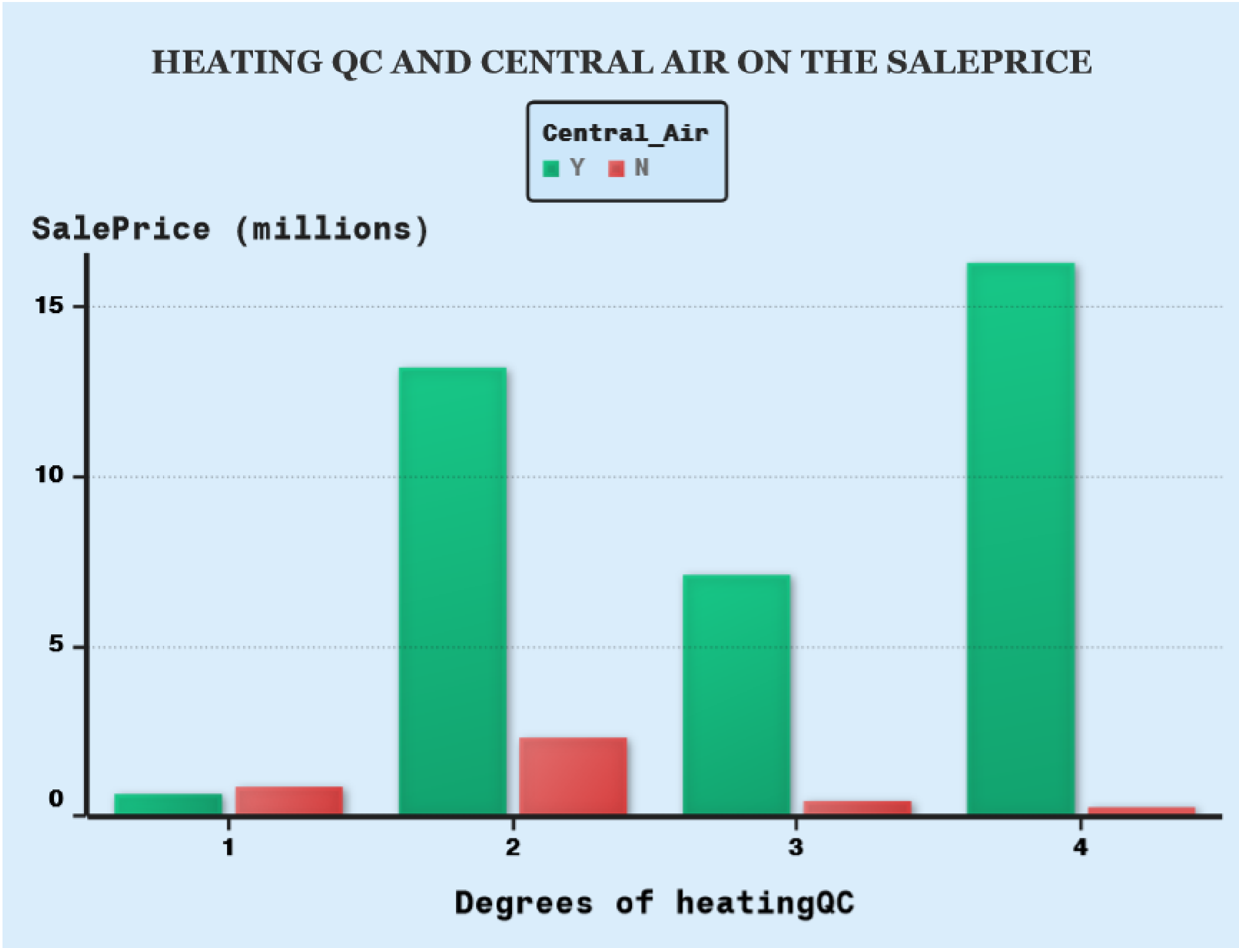
- Homes with excellent heating quality condition have the highest sum of sales price(16.6 million).
- Homes with typical average heating quality condition have slightly lower sum of sales price than excellent ones(15.5 million).
- Homes with good heating quality condition have a lower sum of sales price at around 7.6 million.
- Homes with fair heating quality condition have the lowest sum of sales price, at around 1.5 million.



The bar plot shows sales price of homes, by Central Air conditioning

It is clear that houses with Central Air conditioning have higher sum of sale prices than those without

- Central air conditioning provides consistent cooling throughout the home, enhancing comfort, especially in warmer climates or during hot seasons. Buyers often prioritize this feature, leading to increased demand and higher prices for homes equipped with central air conditioning.

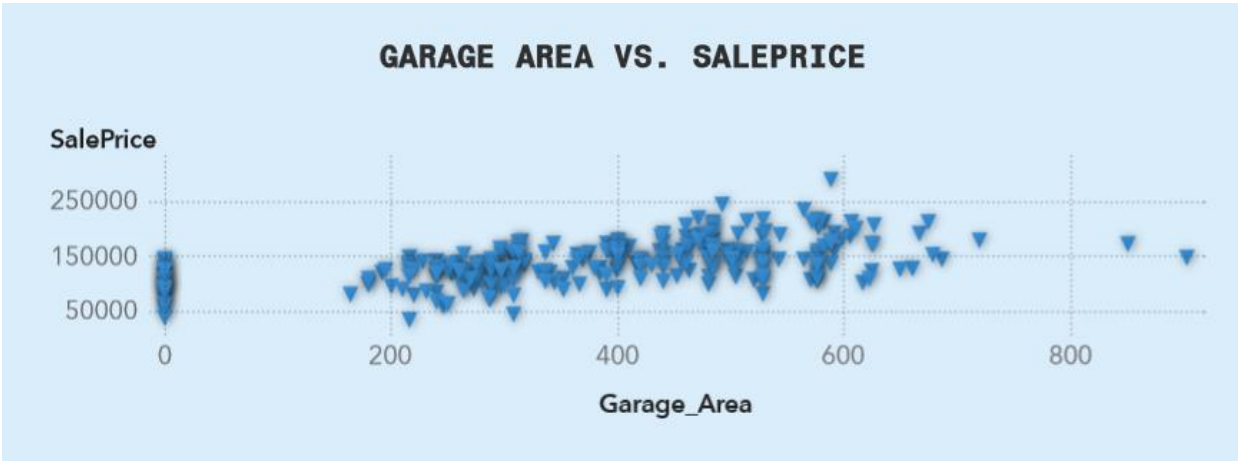


The graph shows the relationship between the sale price of homes (in millions) and the number of degrees of heating and Central Air conditioning the home has.

As we can see from the graph, there is a positive correlation between the sale price of a home and the number of degrees of heating and Central Air conditioning the home has.

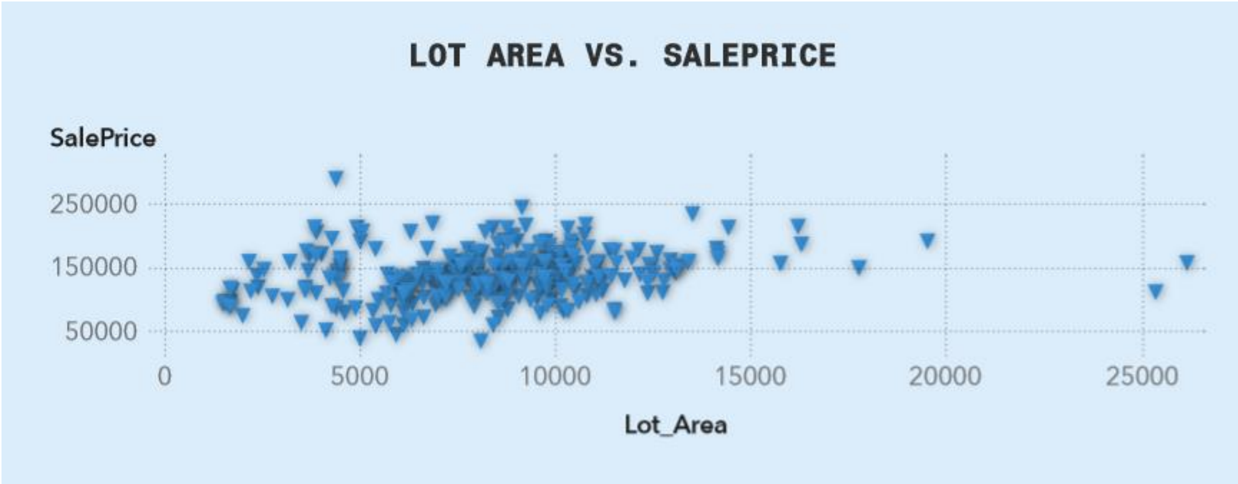
This means that homes with more heating and cooling tend to sell for more money as they are more comfortable to live in, which can make them more appealing to buyers.

GARAGE AREA AND LOT AREA ON HOUSE PRICE



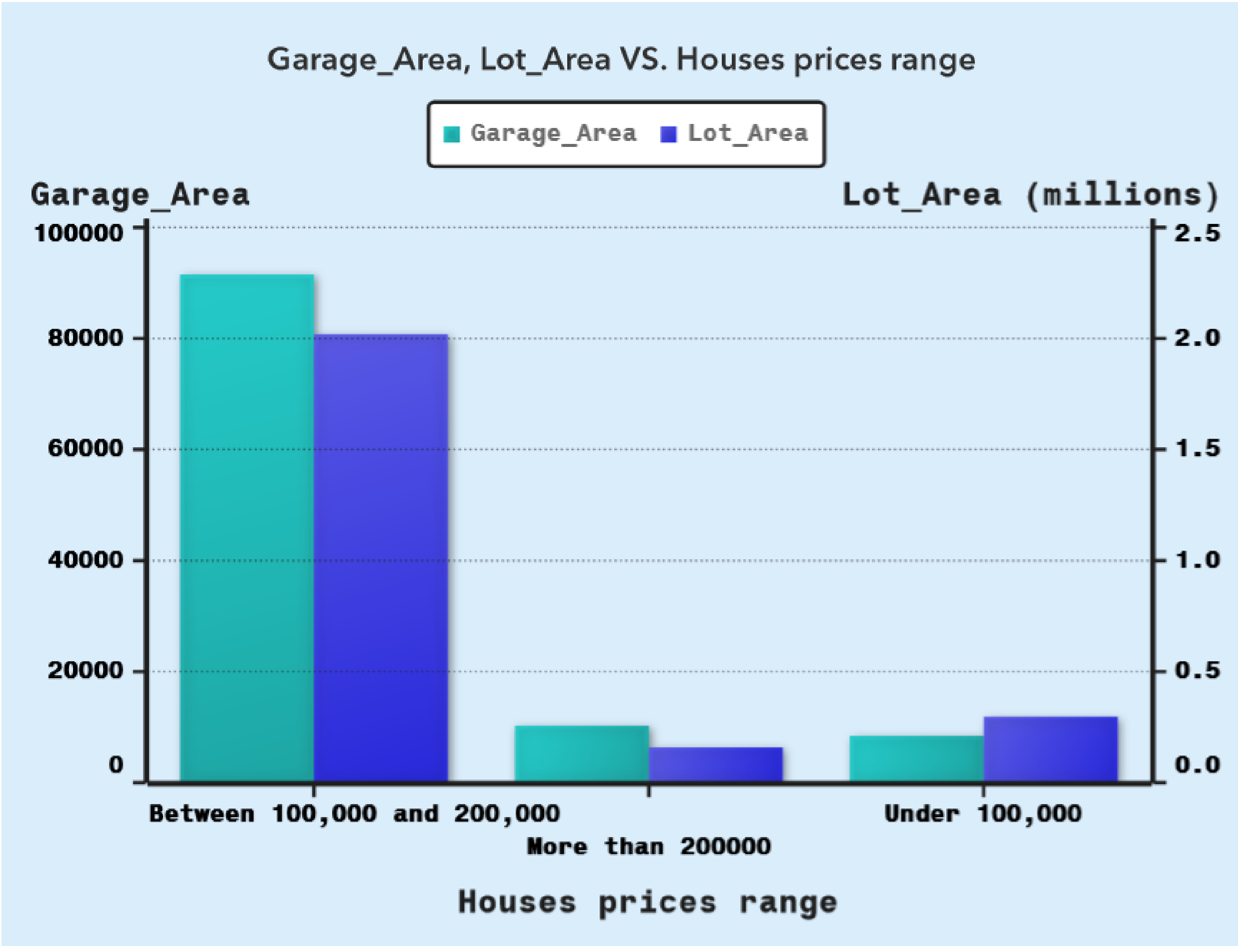
The scatter plot shows the relationship between the size of a garage and the sale price of the house it's attached to.

It suggests that there is a positive correlation between the size of a garage and the sale price of a house. This means that houses with larger garages tend to sell for more money than houses with smaller garages.



The scatter plot shows relationship between lot area and sale price for houses.
The graph shows that there is a positive correlation between lot area and sale price. This means that houses with larger lots tend to sell for more money than houses with smaller lots. There are a few possible explanations for this.

- A larger lot gives homeowners more space to enjoy, such as for a larger yard, a pool, or a garden. This can make the house more desirable to buyers and willing to pay more for it.
- A larger lot may also give homeowners the potential to develop the property further, such as subdividing the lot. This can also make the house more valuable.
- In some cases, larger lots may be located in more desirable areas, which can also contribute to a higher sale price.

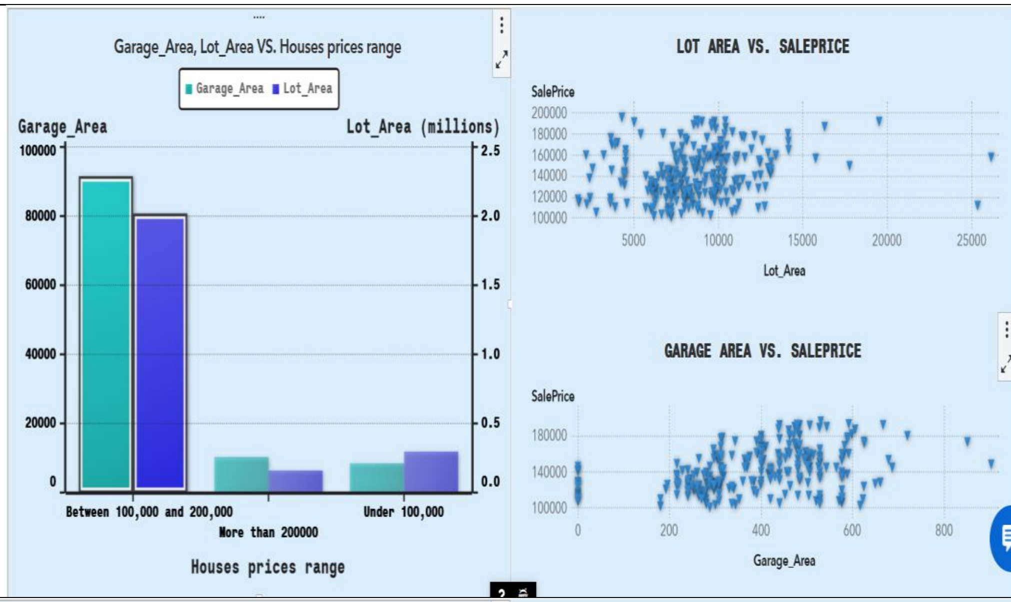
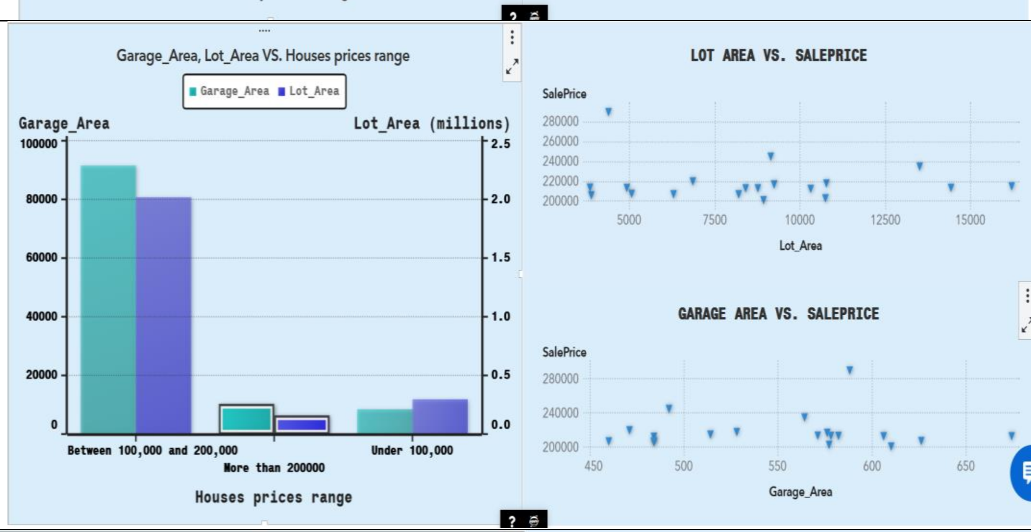
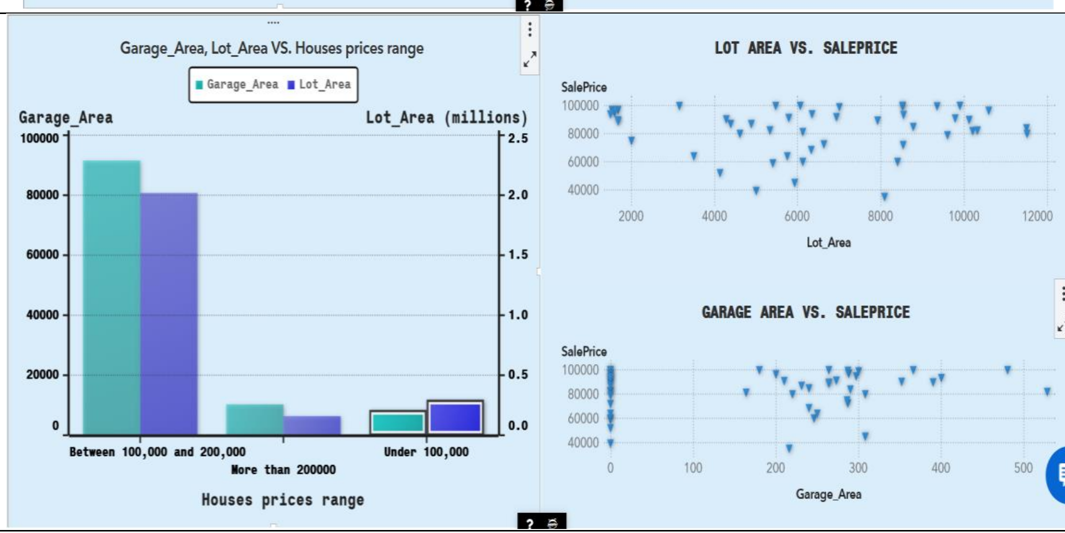


From that plot we could deduce that houses their prices range between 100,000 and 200,000, have a highest summation of garage area and lot_area.

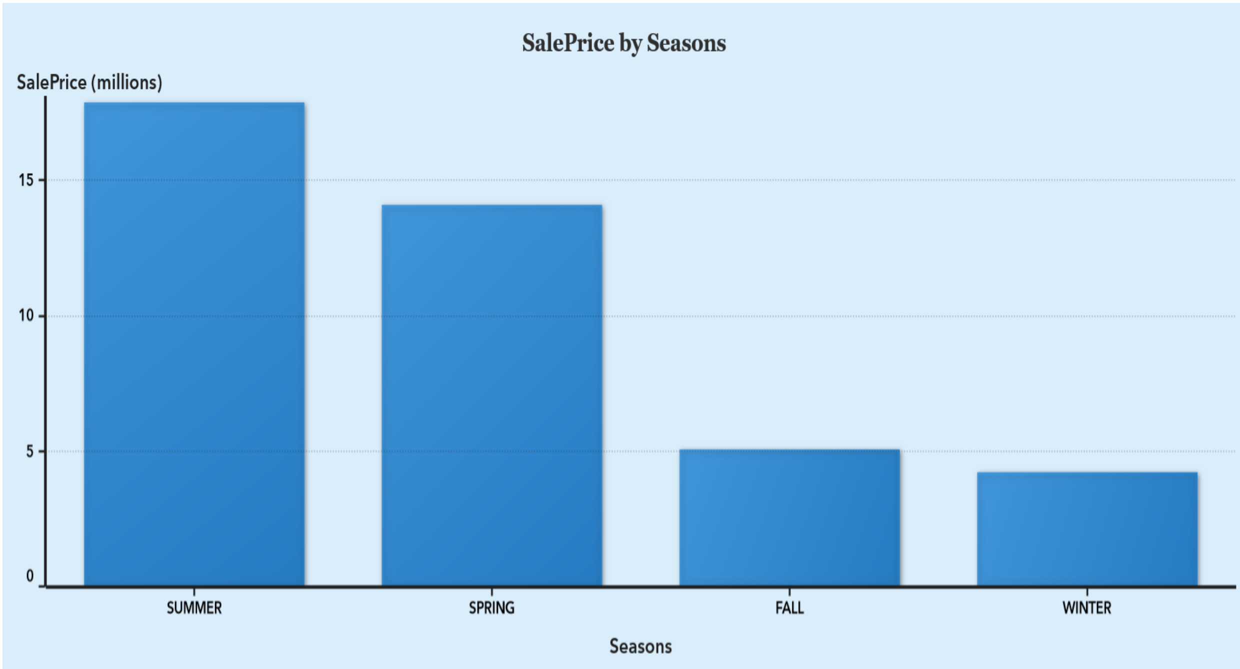
This ensures the positive correlation between both the size of a garage and lot area with the sale price of a house

Also for houses more than 200,000 , the Garage Area seems to be more important than lot area.

Here is the distribution of both the size of a garage and lot area on sale price ranges

Between 100,000 and 200,000	
More than 200,000	
Under 100,000	

SEASONS AND SALEPRICE



Regarding the bar plot between sale price for houses by season :

- The summation of sale price for houses in the summer is the highest.
- The summation of sale price for houses in the winter is the lowest.
- The summation of sale price for houses in the spring is higher than the sale price for houses in the fall

Here are some possible explanations for that:

➢ Pleasant weather: Spring and summer are more attractive times to buy a house with pleasant weather for outdoor activities like house viewings and landscaping.

- Schooling: Families with children might prefer to move during the summer to minimize disruption to their schooling.

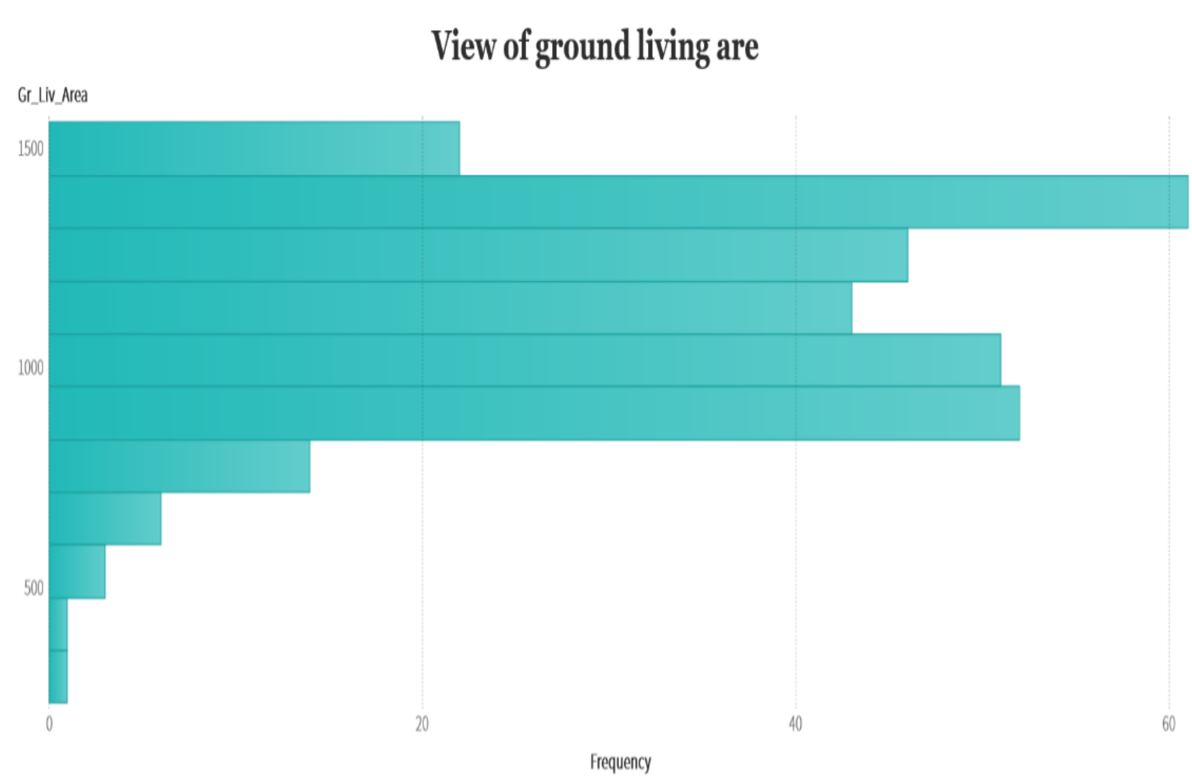
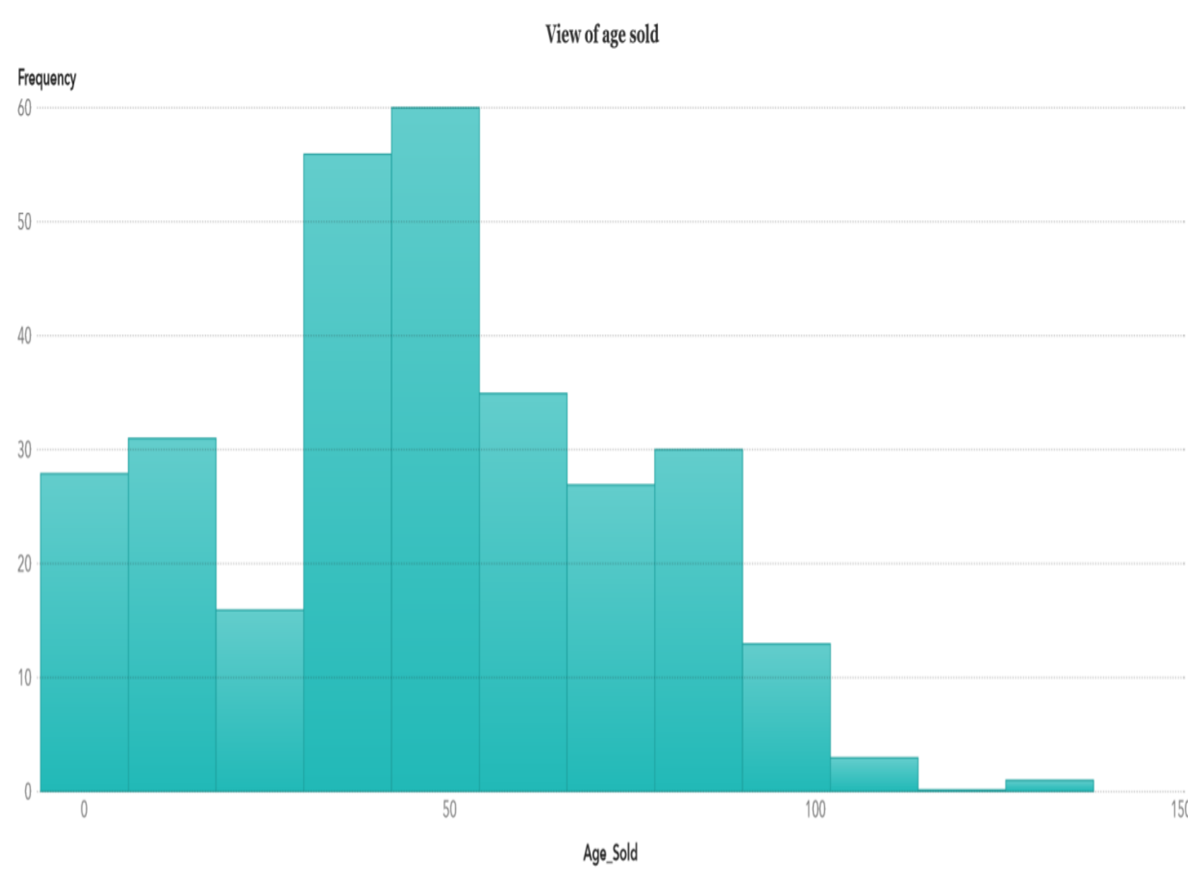
Report phase

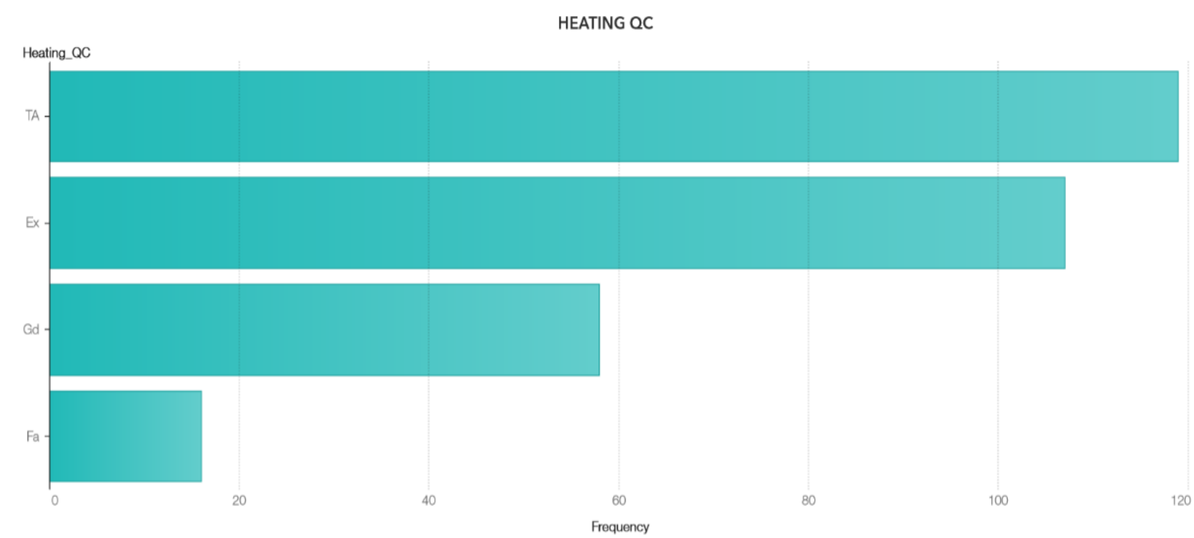
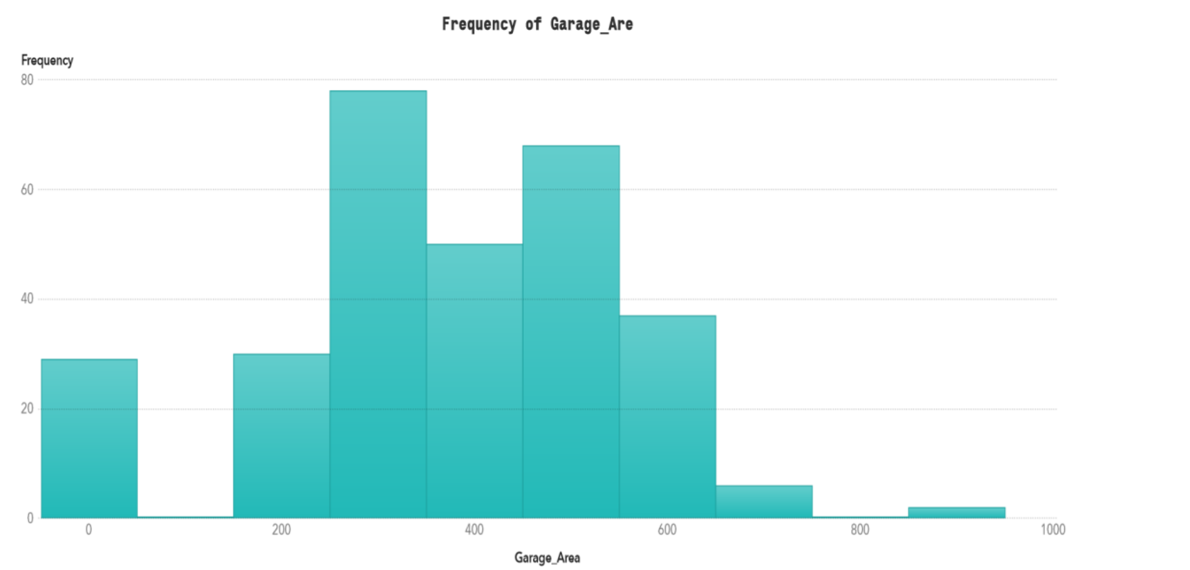
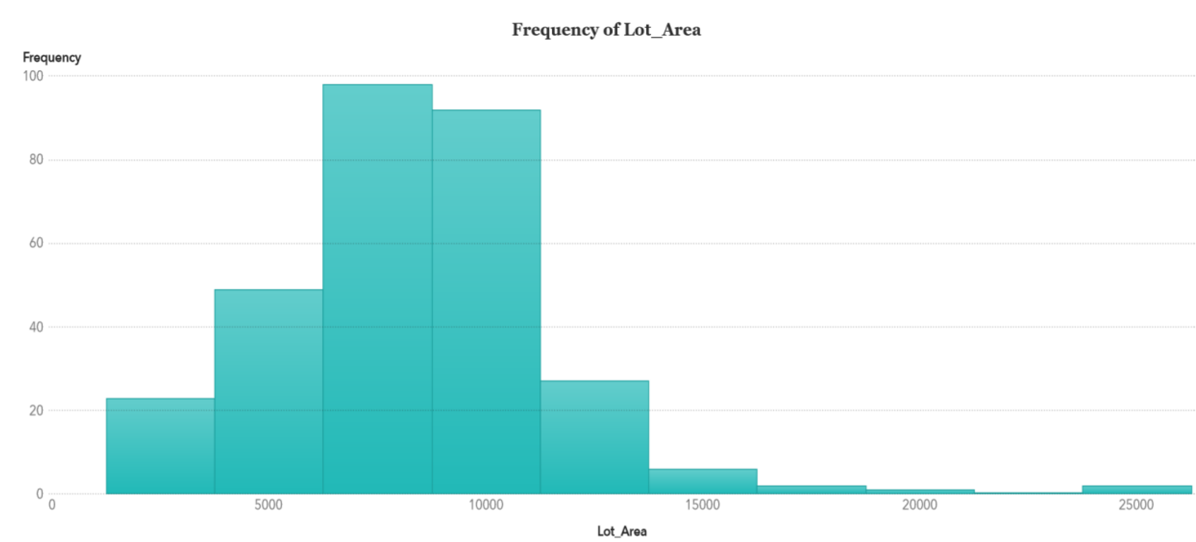
In the Report phase, we develop interactive reports that can be shared via the web or a mobile device.

- Hidden pages

Hidden pages are a feature of SAS Visual Analytics that allow us to include a data item in the query results when we want its values, but do not want it to be displayed

Regarding our report we have made 5 hidden pages to enable viewers to see additional details.

Ground Living Area	
Age Sold	

Heating QC	
Garage Area	
Lot Area	

My Contribution

1- Prepare Phase:

During the Preparation phase, I rectify any data quality issues and generate any necessary calculated items for analysis.

• I inferred that we can introduce a new data category from the sale price, which will have three classes: under 100,000, between 100,000 and 200,000, and over 200,000.

• I determined that we can modify the measure (season_sold) from numerical values (1,2,3,4) to corresponding seasons (winter, spring, summer, fall).

• I can transform the heating quality condition from categorical values (ex, fa, etc.) to numerical equivalents.

2- GARAGE AREA AND LOT AREA ON HOUSE PRICE

In My study of house prices, I utilized scatter plots to illustrate the relationships between the size of a garage, the lot area, and the sale price of houses.

My analysis suggests a positive correlation between the size of a garage and the sale price of a house, indicating that houses with larger garages generally command higher prices than those with smaller garages.

Similarly, I found a positive correlation between lot area and sale price. Houses with larger lots tend to fetch higher prices for several reasons:

• A larger lot provides homeowners with more space for amenities like a spacious yard, a pool, or a garden, increasing the property's appeal to buyers.

• A larger lot also offers potential for further property development, such as lot subdivision, enhancing the property's value.

• In some instances, larger lots are situated in more desirable locations, contributing to a higher sale price.

Using a histogram plot, I observed that houses priced between 100,000 and 200,000 have the highest combined garage area and lot area, reinforcing the positive correlation between these factors and the sale price of a house.

For houses priced above 200,000, the garage area appears to have a greater impact on the sale price than the lot area. This distribution of garage size and lot area across different price ranges further underscores their influence on house prices.

3- Report phase:

During the Report phase, I create interactive reports that can be disseminated through web or mobile platforms.

• Hidden Pages: Hidden pages are a unique feature of SAS Visual Analytics. They allow us to incorporate a data item in the query results for its values, without displaying it.

In our report, i have utilized this feature to create 5 hidden pages. These hidden pages serve as a tool for viewers to access more detailed information without cluttering the main report.

Knowledge Attained from training and Project

1. **Understanding Theory:**

   The coursework cultivated an in-depth comprehension of SAS Visual Analytics, its methodologies, and its practical application to real-world datasets, encompassing the five phases: Access, Investigate, Prepare, Analyze, and Report.

2. **Mastering Technical Proficiencies:**

   The learning journey enhanced my proficiency in utilizing SAS Visual Analytics, which includes skills in data access, preparation, analysis, and report generation.

3. **Practical Data Analysis:**

   The project offered hands-on experience in analyzing a dataset using SAS Visual Analytics, including the identification of patterns, relationships, and trends in the data.

4. **Visualizing Data:**

   The learning process imparted knowledge on how to visualize data in a meaningful way, including the use of different types of charts and graphs to represent data and uncover hidden relationships.

5. **Preparing Data for Analysis:**

   The project aided in understanding how to prepare data for analysis, including skills in cleaning data, dealing with missing values, and creating new data categories.

6. **Creating Interactive Reports:**

   The project provided practical experience in creating interactive reports that can be shared via the web or a mobile device, including understanding how to use hidden pages to include data items in the query results without displaying them.

7. **Gaining Domain Knowledge:**

   The project offered practical insights into the housing market, including the factors that influence house prices. This includes understanding variables such as lot area, house style, heating quality, central air, ground living area, number of fireplaces, garage area, sale price, age when sold, and the season when the house was sold.

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Thanks